Jack Luk

jackluk518@gmail.com | linkedin.com/in/jack-luk | jackluk.com | Vancouver, BC

TECHNICAL SKILLS

Languages: C++, Embedded C, Python, VHDL, MATLAB, Bash
Embedded & Hardware: FPGA (Intel Cyclone V), STM32, Arduino, CAN bus, sensor and servo interfacing, soldering
Lab & Debug: Oscilloscopes, bench power supplies, ModelSim simulation, Quartus Prime synthesis
Edge AI & Vision: OpenCV, MediaPipe, scikit-learn, Raspberry Pi 5, NVIDIA Jetson
Software & Tools: Linux, Git, Docker, NumPy, pandas, SolidWorks

PROJECTS

Vision-Based Rear Monitor
Oct 2026 – Present

Python, OpenCV, Embedded C, CAN bus, Raspberry Pi 5, STM32

- Developing a **two-node embedded system** for cyclists: a Raspberry Pi 5 perception node detects vehicles approaching from behind and transmits threat levels over **CAN bus** to an **STM32** alert unit driving a light and buzzer.
- Architected the CAN message set (detection status, **100 ms heartbeat**) with a **500 ms timeout failsafe** that drives a fault pattern when communication is lost, mirroring automotive ECU practice.

FPGA Digital Alarm Clock
Jul 2026 – Aug 2026

VHDL, Intel DE10-Standard (Cyclone V), Quartus Prime

- Built a 24-hour **alarm clock from pure logic** (no processor or OS) in a team of 2: an FSM drives **3 modes** (run, set time, set alarm) on six 7-segment displays, verified on a Cyclone V FPGA.
- Divided the **50 MHz** board clock to a **1 Hz** tick; when switch bounce made one press skip the time ahead, fixed it with **debounced, edge-detected** inputs so each press registers exactly once.
- Integrated reusable **BCD counters** (0–59, 0–23), editing and display subsystems through a shared control interface, catching bugs early with **ModelSim testbenches**.

Automatic Blueberry Sorter
Sep 2025 – Dec 2025

Arduino, TCS3200 Sensor, SG90 Servo, DC Motors

- **Led a team of 8 as project manager** to automate the slow hand-sorting small berry growers and packers do, delivering a proof-of-concept within a 4-month term.
- Programmed Arduino firmware that checks **TCS3200** colour readings against a calibrated RGB range and sweeps an SG90 servo's **30° flap** to reject bad berries and debris.
- Engineered a 12 V **centrifugal mechanism** that sorts berries into **2 size classes** and comes apart for washing.

AI Posture Detection System
Nov 2022 – Apr 2023

Python, OpenCV, MediaPipe, scikit-learn

- Built a webcam app that warns people when they **slouch at a desk**, needing no extra hardware: it classifies posture from **33 MediaPipe landmarks**, trained on **~1,500 self-labelled images**.
- Compared **4 classifiers** (best **88.5%** on test data) and diagnosed the gap between test and live accuracy as a too-uniform dataset. Advanced to the **Greater Vancouver Regional Science Fair**.

EXPERIENCE & LEADERSHIP

Tech Team Member, *SFU Racerbot Club* — Oct 2026 – Present

- Set up and maintain the **NVIDIA Jetson** compute platform (Ubuntu, JetPack) of a **1/10-scale autonomous race car**.
- Help design and test **mechanical and electrical** hardware improvements, and keep the car's physical components running.

Youth Leader, *33rd Richmond Scouts (Richmond, BC)* — Sep 2020 – Present

- Coordinate weekly programs and lead large-scale fairs and camps across **6 years** of continuous volunteer leadership.

First Year Lead, *SFU Canadianized Asian Club (Burnaby, BC)* — Oct 2025 – Aug 2026

- Organized events and fundraisers that engaged SFU's Asian student community.

Private STEM Tutor, *STEM Power to Learn (Remote)* — Oct 2023 – Jun 2025

- Planned and delivered weekly online math sessions for elementary students over **2 academic years**.

EDUCATION

Simon Fraser University, Burnaby, BC — Anticipated May 2029

Bachelor of Applied Science, Systems Engineering

Relevant coursework: Electric Circuits, C++ Programming, Digital Logic Design (VHDL), Engineering Graphics and CAD